

lab3

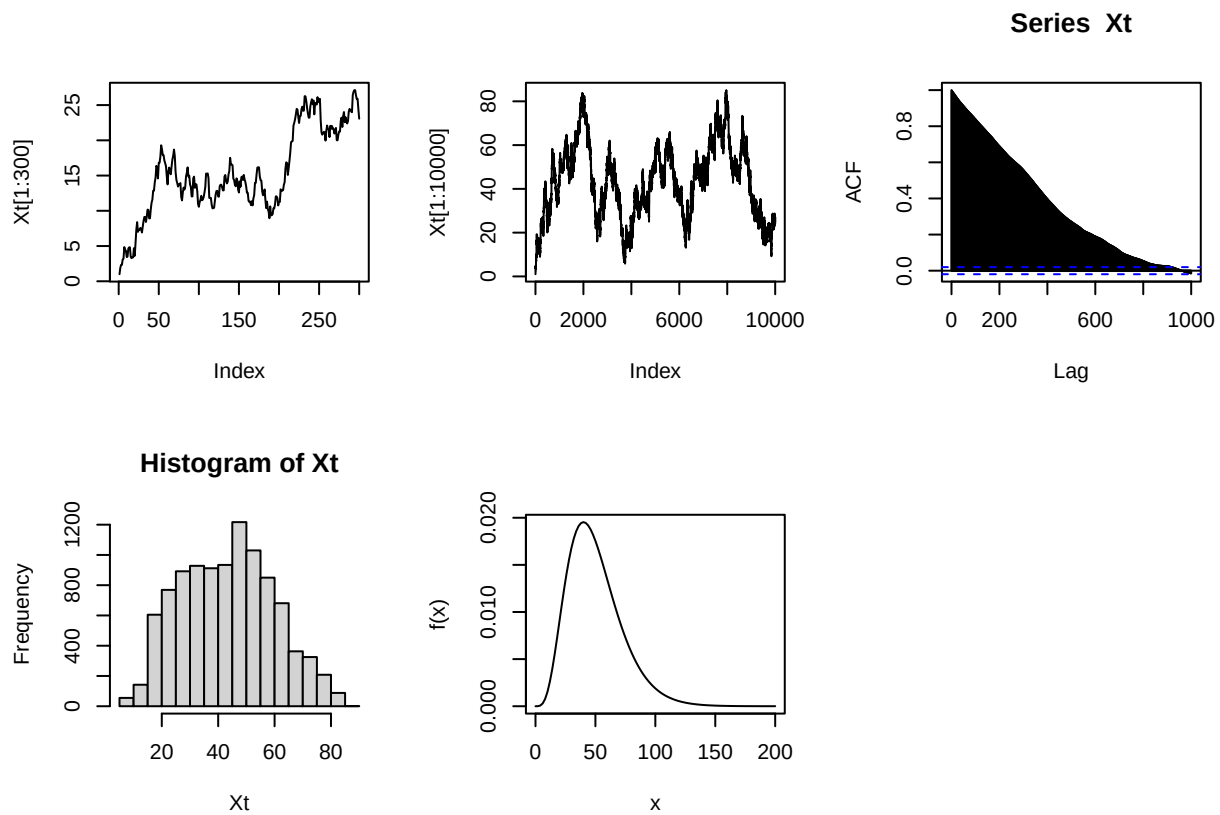
2026-02-05

Q1

a

```
## [1] 0.9814563
```

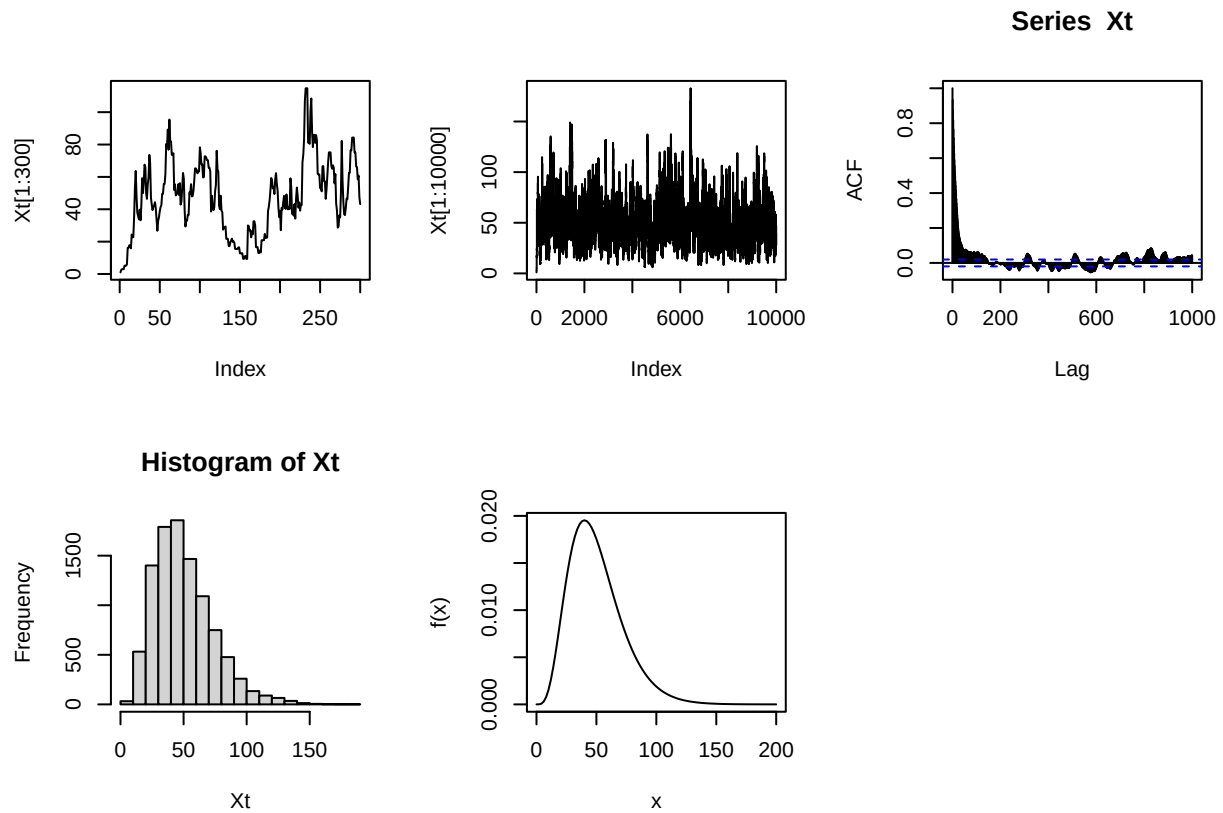
We can see in the ACF plot that we have not reached a markov chain.



The chain seems to need a burn-in as it moves far away from the initial value, looking at the density plot $x=1$ is part of a low probability region. X_t spans from 0 to 20, looking at the density the MCMC chain hasn't explored the density at all. We still have a lot of autocorrelation as well, judging from the trace plot it is because the step size is too small. The acceptance rate is very high, 98%.

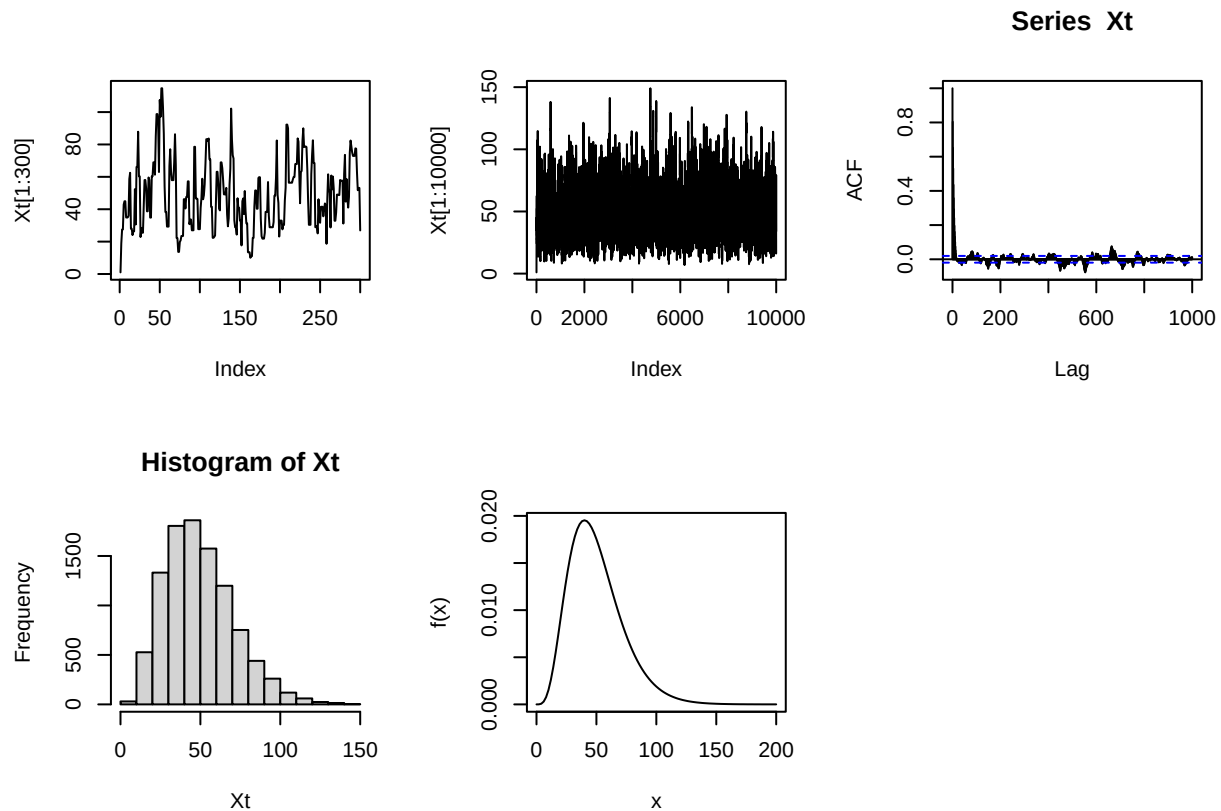
b

```
## [1] 0.8527184
```



The histogram looks very good. Lets try a distribution with a larger stepsize, normal with larger variance, $X \sim N(\sigma = 20, \mu = 0)$.

```
## [1] 0.7108738
```



c + d

This looks as good as in b), we have lower acceptance rate but the correlation drops faster. In a) we had too low of a step size. In c) we more effectively traverse the distribution. The chi-squared proposal still seems to be the best in terms of the speed of convergence. The chi-square proposal is state-dependent, e.g. $Y \mid X_t = x \sim \chi^2_{[x]+1}$, so its typical proposal size changes with the current location of the chain. This often yields larger, more effective moves when the chain is in higher-density regions away from 0, so the chain can reach and traverse the typical set in fewer iterations than a fixed-scale random-walk. Both b) and c) got good histograms.

e + f

```
## [1] "Part a:"
```

```
## Mean: 43.37288
```

```
## Median: 43.6496
```

```
## [1] "Part b:"
```

```
## Mean: 50.26692
```

```
## Median: 46.58431
```

```
## [1] "Part c:"
```

```
## Mean: 49.58739
```

```
## Median: 47.00557
```

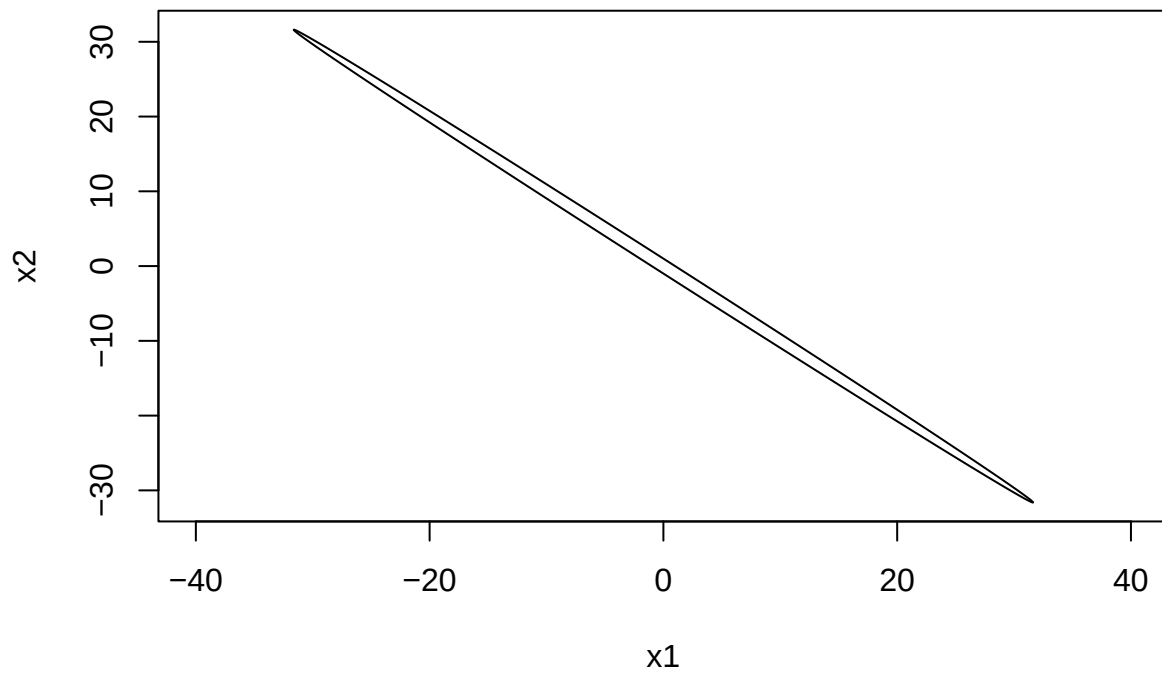
Theoretical values are:

$$f(x) \propto x^{\alpha-1} \cdot e^{-x/\theta} \Rightarrow E[X] = \alpha\theta$$

Our mean is therefore $5 \cdot 10 = 50$. Part b was closest and part c was also close.

Q2

a



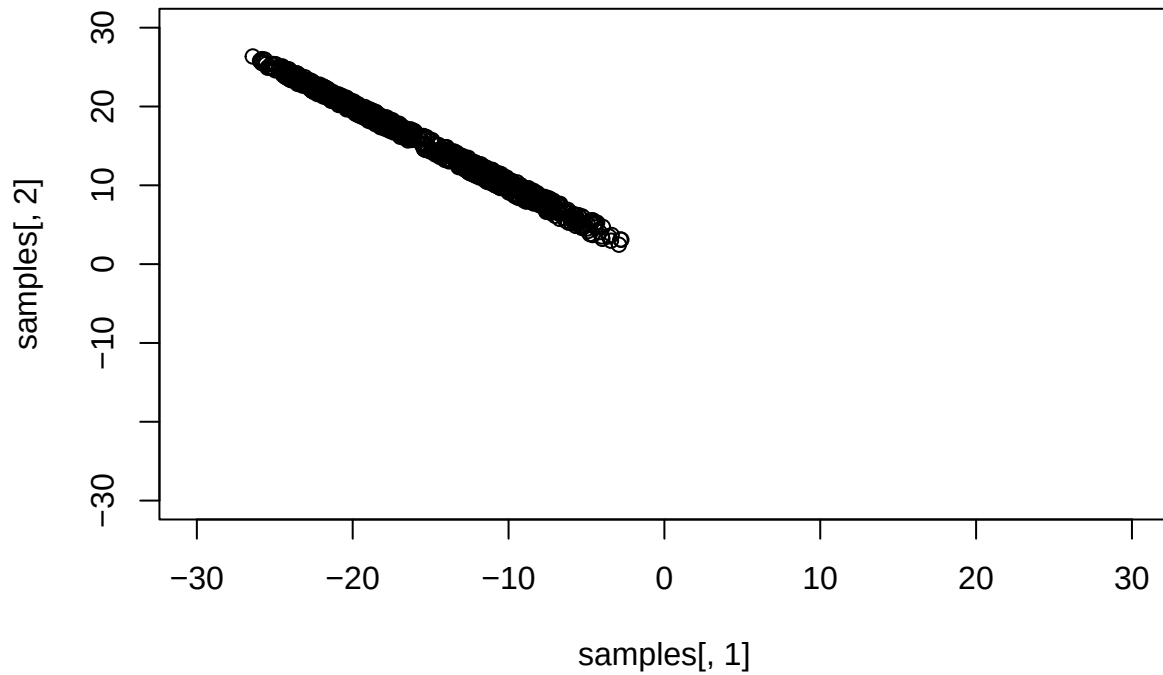
b

The conditional distribution of $x_1|x_2$ will still be uniform but on the interval:

$$x_2 \in -\frac{wx_1}{2} \pm \sqrt{\frac{(wx_1)^2}{4} - x_1^2 + 1}$$

The problem is symmetrical for $c_2|x_1$.

c + d



```
## [1] 0
```

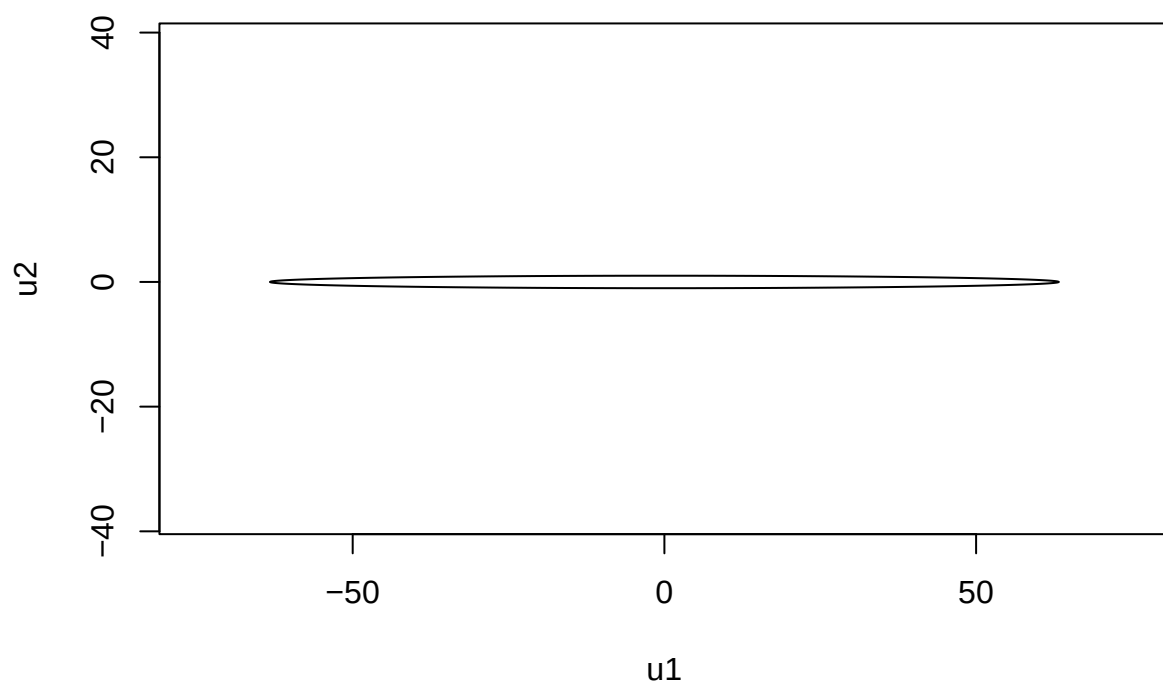
The true result would be 50%. The whole ellipse doesn't seem to be explored, this makes sense as the ellipse is so thin, x_1 and x_2 moves so slowly through the ellipse, so if the initial value for x_1 is not from the target distribution it will require a long burnin for the final sample to be.

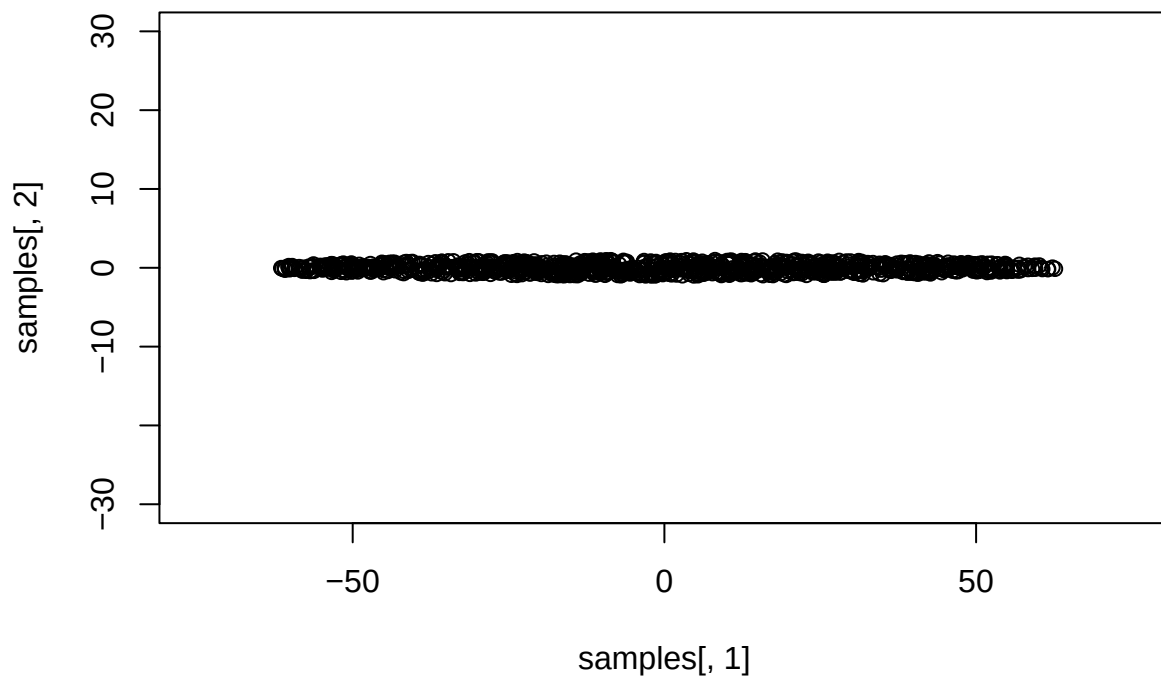
e

We get the following boundary:

$$4 = (2 + w) \cdot U_1^2 + (2 - w) \cdot U_1^2$$

If we scale the U_1 and U_2 axis by $\sqrt{2+w}$ and $\sqrt{2-w}$ we see we have a circle. Therefore we have an ellipse here. As w is close to 2 the ellipse is still stretched a lot, the difference in the axes is a factor of 44.7.





```
## [1] 0.498
```

Here we converge way faster. For part c we were not even close to 50%. The variance for this estimate seems to be large. # Appendix

```
f <- function(x) {

  result <- x^4*exp(-x/10)/(2.4*10^6)
  result[x<0] <- 0
  return(result)
}

n <- 10000
burnin <- 300

Xt <- c(1)

i <- 1
acc <- 0

set.seed(543)
while (i < (n + burnin)) {
  U <- runif(1)
```

```

x_star <- rnorm(1, Xt[i], 1)

MH1 <- f(x_star) / f(Xt[i])
MH2 <- dnorm(Xt[i], mean = x_star) / dnorm(x_star, mean = Xt[i])

MH <- MH1 * MH2

if (U < MH) {
  acc <- acc + 1
  Xt[i + 1] <- x_star
} else {
  Xt[i + 1] <- Xt[i]
}

i <- i + 1
}

acc_rate <- acc / i

print(acc_rate)

par(mfrow = c(2, 3))
plot(Xt[1:300], t = "l")
plot(Xt[1:10000], t = "l")

Xt <- Xt[-c(1:burnin)]
XtA <- Xt

acf(Xt, 1000)

hist(Xt)
curve(f(x), 0, 200)

Xt <- c(1)

i <- 1
acc <- 0

set.seed(543)
while (i < (n + burnin)) {
  U <- runif(1)
  x_star <- rchisq(1, floor(Xt[i] + 1))

  MH1 <- f(x_star) / f(Xt[i])
  MH2 <- dchisq(Xt[i], df = floor(x_star + 1)) / dchisq(x_star, df = floor(Xt[i] + 1))

```



```

MH <- MH1 * MH2

if (U < MH) {
  acc <- acc + 1
  Xt[i + 1] <- x_star
} else {
  Xt[i + 1] <- Xt[i]
}

i <- i + 1
}

acc_rate <- acc / i

print(acc_rate)

par(mfrow = c(2, 3))
plot(Xt[1:300], t = "l")
plot(Xt[1:10000], t = "l")

Xt <- Xt[-c(1:burnin)]
XtB <- Xt

acf(Xt, 1000)

hist(Xt)
curve(f(x), 0, 200)

Xt <- c(1)

i <- 1
acc <- 0

sd <- 20

set.seed(543)
while (i < (n + burnin)) {
  U <- runif(1)
  x_star <- rnorm(1, Xt[i], sd)

  MH1 <- f(x_star) / f(Xt[i])
  MH2 <- dnorm(Xt[i], mean = x_star, sd = sd) / dnorm(x_star, mean = Xt[i], sd = sd)

  MH <- MH1 * MH2

```

```

if (U < MH) {
  acc <- acc + 1
  Xt[i + 1] <- x_star
} else {
  Xt[i + 1] <- Xt[i]
}

i <- i + 1
}

acc_rate <- acc / i

print(acc_rate)

par(mfrow = c(2, 3))
plot(Xt[1:300], t = "l")
plot(Xt[1:10000], t = "l")

Xt <- Xt[-c(1:burnin)]
XtC <- Xt

acf(Xt, 1000)

hist(Xt)
curve(f(x), 0, 200)

print("Part a:")
cat("Mean: ", mean(XtA), "\n")
cat("Median: ", median(XtA), "\n")

print("Part b:")
cat("Mean: ", mean(XtB), "\n")
cat("Median: ", median(XtB), "\n")

print("Part c:")
cat("Mean: ", mean(XtC), "\n")
cat("Median: ", median(XtC), "\n")

w <- 1.999

f_x2 <- function(x1, sign = 1) {
  -w*x1/2 + sign * sqrt((w*x1/2)^2 - x1^2 + 1)
}

```

```

}

x1 <- seq(-40, 40, by=0.01)
x2 <- c(f_x2(x1, 1), f_x2(x1, -1))
x1 <- c(x1, x1)

plot(x1, x2, type = "l")

n <- 1000
burnin <- 300
samples <- matrix(NA, nrow=n + burnin, ncol=2)

set.seed(123456)
x1 <- 0
for (i in 1:(n+burnin)) {

  if (i == 1) {
    x1 <- 0
  } else {
    min_x1 <- f_x2(x2, -1)
    max_x1 <- f_x2(x2, 1)
    x1 <- runif(1, min_x1, max_x1)
  }

  min_x2 <- f_x2(x1, -1)
  max_x2 <- f_x2(x1, 1)
  x2 <- runif(1, min_x2, max_x2)

  samples[i, ] <- c(x1, x2)
}

samples <- samples[-(1:300), ]

plot(samples[, 1], samples[, 2], xlim = c(-30, 30), ylim = c(-30, 30))

print(mean(samples[, 1] > 0))

w <- 1.999

f_u1 <- function(u2, sign = 1) {
  sign*sqrt((4 - (2+w)*u2^2) / (2-w))
}

f_u2 <- function(u1, sign = 1) {
  sign*sqrt((4 - (2-w)*u1^2) / (2+w))
}

u1 <- seq(-75, 75, by = 0.01)
u2_top <- f_u2(u1, +1)
u2_bot <- f_u2(u1, -1)

```

```

plot(u1, u2_top, type = "l", asp = 1, xlab = "u1", ylab = "u2")
lines(u1, u2_bot)

n <- 1000
burnin <- 300
samples <- matrix(NA, nrow=n + burnin, ncol=2)

set.seed(1234567)
u1 <- 0
for (i in 1:(n+burnin)) {

  if (i == 1) {
    u1 <- 0
  } else {
    min_u1 <- f_u1(u2, -1)
    max_u1 <- f_u1(u2, 1)
    u1 <- runif(1, min_u1, max_u1)
  }

  min_u2 <- f_u2(u1, -1)
  max_u2 <- f_u2(u1, 1)
  u2 <- runif(1, min_u2, max_u2)

  samples[i, ] <- c(u1, u2)
}

samples <- samples[-(1:300), ]

plot(samples[, 1], samples[, 2], xlim = c(-75, 75), ylim = c(-30, 30))

print(mean((samples[, 1] + samples[, 2])/2 > 0))

```